

Healthcare Management System Using Agile Methodology

1. Abstract

Healthcare services are very important for society. In many hospitals and clinics, patient records are maintained manually or in separate systems. This leads to problems such as data loss, difficulty in accessing records, and delays in treatment.

A **Healthcare Management System** helps hospitals manage patient information, doctor details, appointments, and medical records digitally. Patients can register, book appointments, and view their reports online. Doctors can access patient history easily, which improves treatment quality.

This project is developed using **Agile methodology**. Agile divides the project into small phases called sprints. Each sprint includes planning, designing, development, testing, and review. Agile allows continuous improvement and easy modification based on user feedback.

The proposed system is user-friendly, secure, and efficient. It reduces paperwork and saves time for both patients and hospital staff. In the future, features such as mobile health apps, online prescriptions, and AI-based diagnosis support can be added.

2. Introduction

2.1 Introduction

A Healthcare Management System is an online system used to manage hospital operations and patient data.

2.2 Problem Identification

- Manual patient record handling
- Time-consuming appointment process
- Difficulty in accessing medical history

2.3 Need of the Project

- Digital patient records
- Faster appointment booking
- Improved healthcare services

2.4 Project Scheduling

The project is divided into multiple sprints using Agile methodology.

2.5 Objectives

- Manage patient and doctor data
- Provide online appointment system
- Improve healthcare efficiency

3. Software Requirement Specification

3.1 Purpose

To develop a healthcare system for managing patient records and hospital services.

3.2 Scope

Used by hospitals, clinics, doctors, and patients.

3.3 Hardware / Software Requirements

Hardware

- Computer / Laptop
- Minimum 4 GB RAM

Software

- Operating System: Windows / Linux
- Programming Language: Java / Python
- Database: MySQL

3.4 Tools

- VS Code / Eclipse
- Web Browser
- Database Server

3.5 Software Process Model

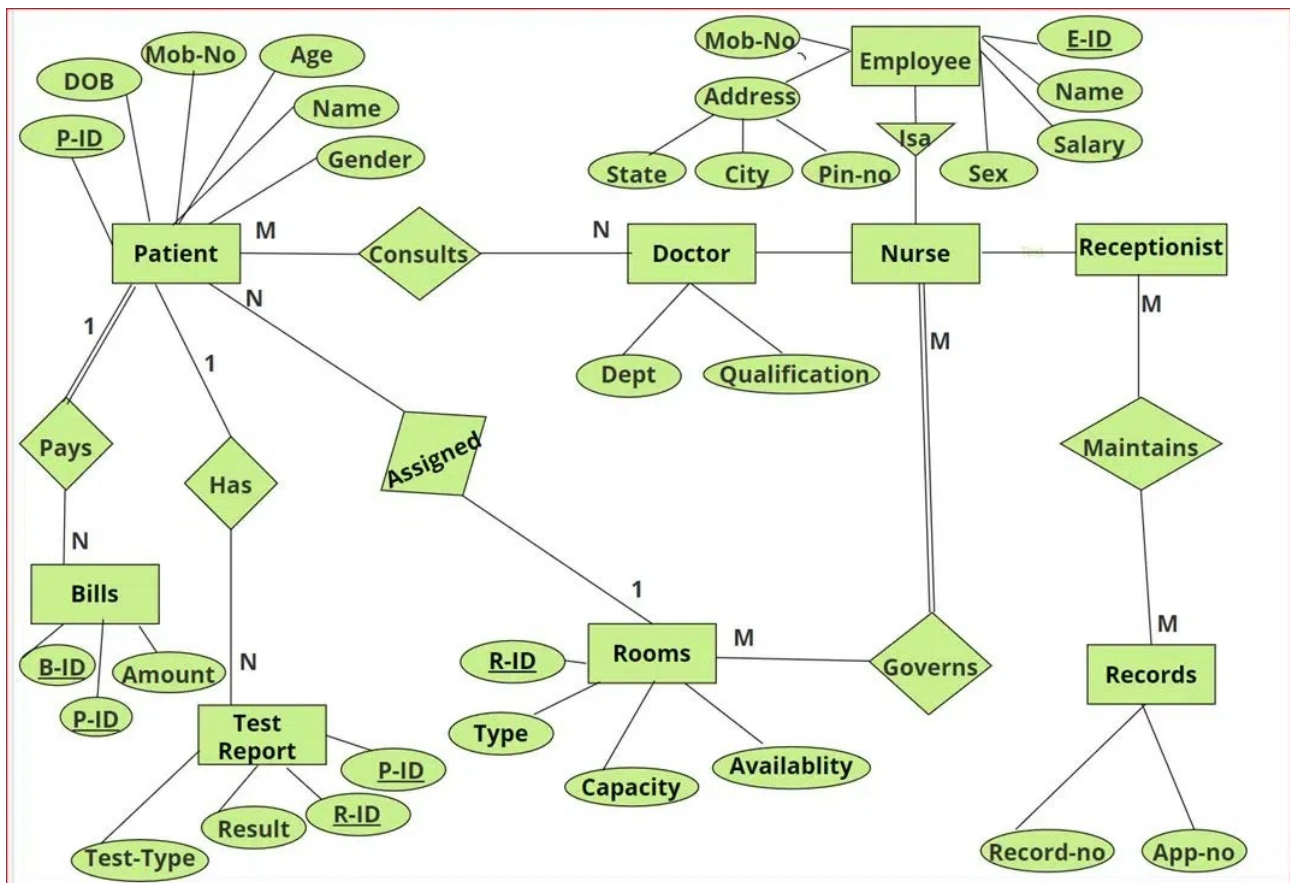
Agile Software Development Model.

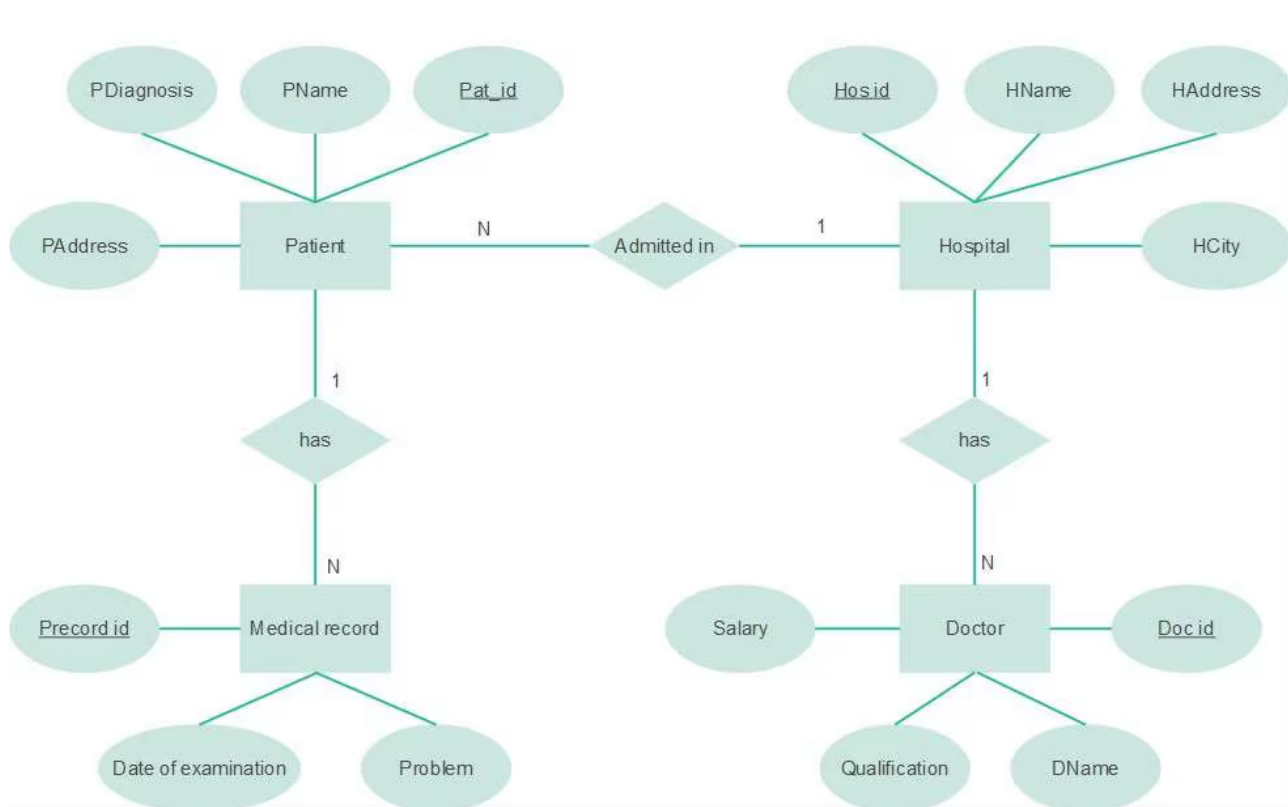
4. System Design

4.1 Data Dictionary

Field Name	Description
PatientID	Unique patient ID
Name	Patient name
Age	Patient age
DoctorID	Doctor identification number
AppointmentDate	Date of appointment

4.2 ER Diagram (REAL)





Entities to draw:

- PATIENT (PatientID, Name, Age)
- DOCTOR (DoctorID, Name, Specialization)
- APPOINTMENT (AppointmentID, Date, Time)

4.3 DFD – Level 0

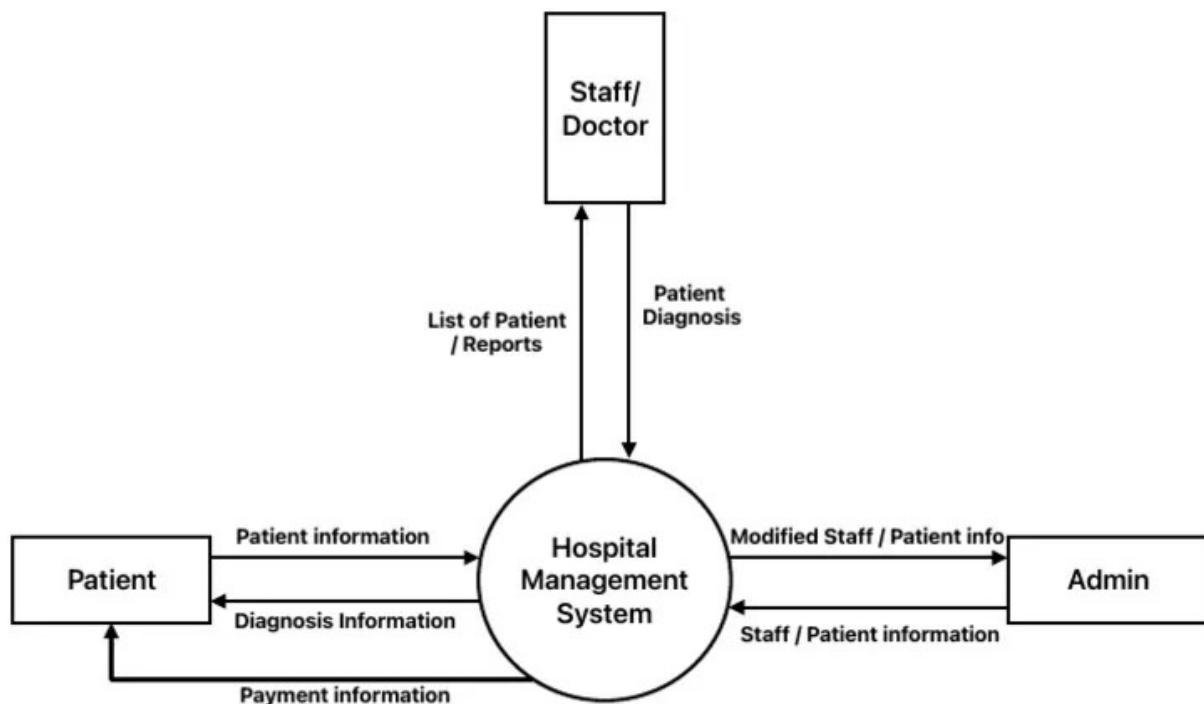


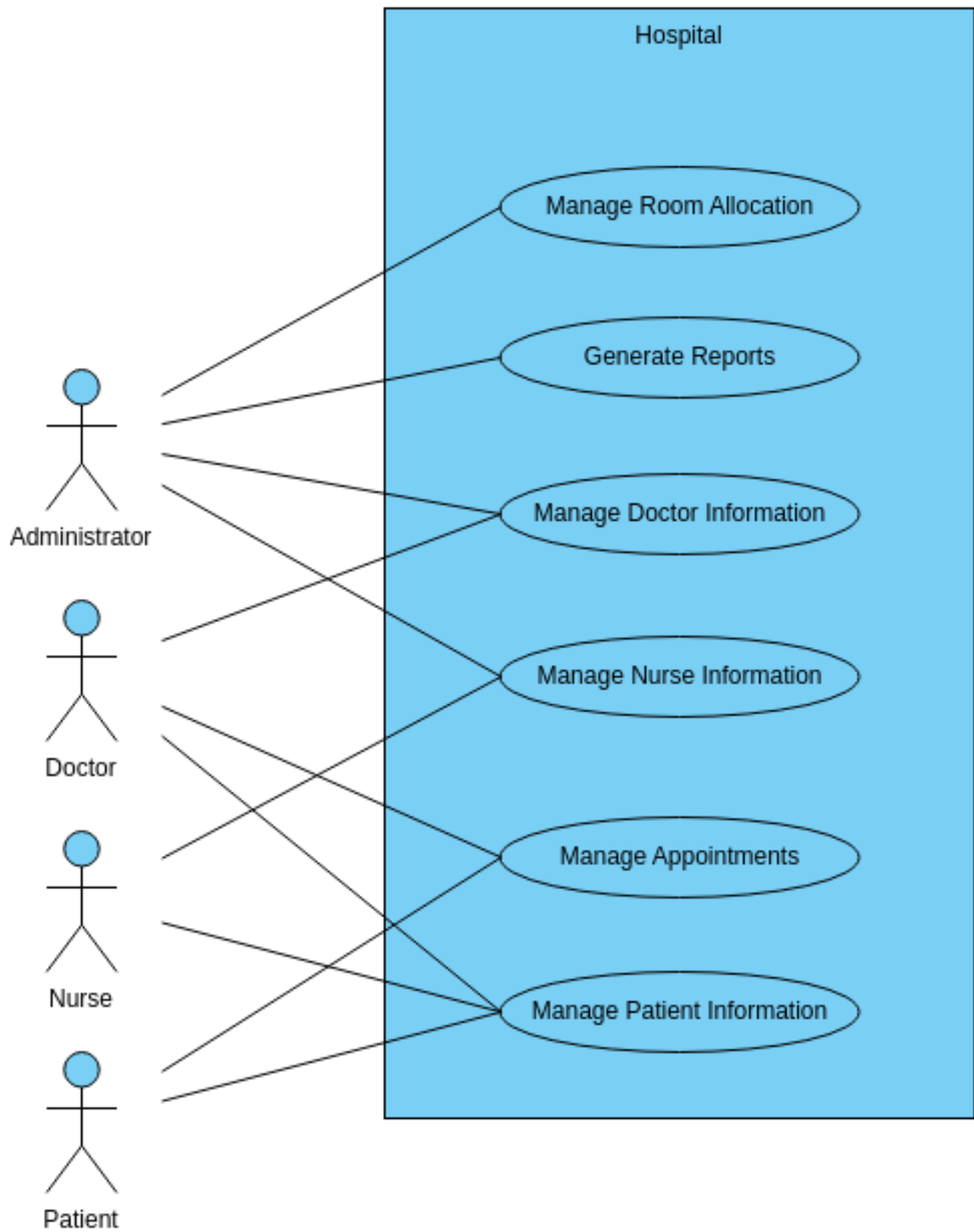
Fig. DFD level 0 of Hospital Management System

Flow:

Patient → Healthcare System → Appointment / Reports

System ↔ Database

4.4 Use Case Diagram (REAL)

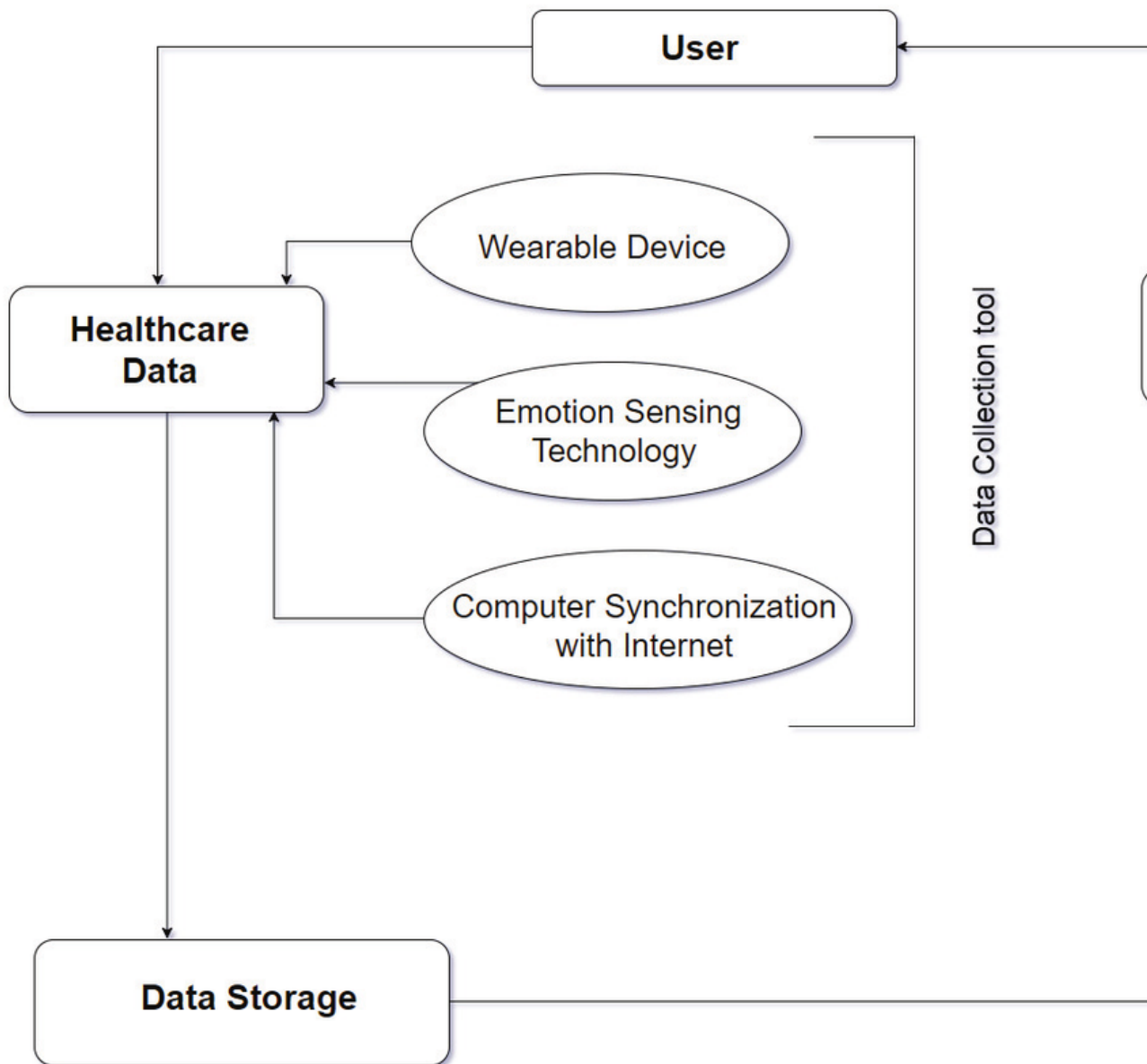


Use Cases:

- Register Patient
- Book Appointment

- View Medical Records
- Update Patient Details

4.5 System Flow Chart



Flow:

Start → Login → Select Service → Process Data → Logout → End

5. Implementation

5.1 Program Code

The healthcare system is implemented using Java/Python with MySQL database.

5.2 Output Screens

- Login Page
- Patient Registration Page
- Appointment Booking Page

6. Testing

6.1 Test Data

- Valid patient details
- Invalid login data

6.2 Test Result

All modules work correctly and produce expected results.

7. User Manual

7.1 How to Use

1. Register and login
2. Book appointment
3. View medical records
4. Logout

7.2 Screen Layout

Simple and easy-to-use interface.

8. Project Applications and Limitations

Applications

- Hospitals
- Clinics
- Healthcare centers

Limitations

- Internet connection required
- Data security risks

9. Conclusion and Future Enhancement

The Healthcare Management System improves hospital efficiency and patient care. Agile methodology ensures flexible and fast development. Future enhancements include mobile health apps, online prescriptions, and AI-based health monitoring.

10. Bibliography & References

- Software Engineering Textbook
- Agile Methodology Notes
- Online Healthcare Technology Resources